

Awesom Living

Sleep Dashboard — Complete Calculation Logic

This document explains, in plain language, how the sleep monitoring dashboard works — what data the GLK bed sensor sends, how it is stored in the database, and how the Dashboard Summary and Detail APIs calculate every number shown on the app screen.

Intended for product verification and client sharing. No code included — only the logic and rules.

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1. What the GLK Bed Sensor Sends

The GLK AI Smart Sleep Monitor is a pad placed under the mattress. It detects a person's presence, heart rate, breathing rate, body movement, and snoring through BCG (ballistocardiography) — measuring tiny vibrations caused by heartbeats and breathing without any wearable device.

The sensor connects to a Raspberry Pi bridge over local WiFi (TCP port 8766). The bridge receives raw data frames and forwards them to our backend server every 5 seconds.

Two types of data frames

Frame Type	Frequency	What It Contains
0x0E — Realtime Vitals	~1 per second	Heart rate (bpm), Respiration rate (breaths/min), Body movement (0-255 raw amplitude), Occupancy status (in bed / out of bed / snoring / light sleep / apnea suspected), Battery level, Signal quality
0x4E — Sleep Stage	1 per minute	Sleep stage classification (awake / light / deep / REM / invalid), Heart rate, Respiration rate, Body movement. NOTE: Currently the device needs configuration to produce per-minute light/deep/REM classifications — this is pending with the GLK team.

Occupancy status codes from 0x0E frame

Code	Status	Meaning
0	Initializing	Sensor is starting up / calibrating
1	In Bed	Person is lying on the pad, breathing normally
2	Apnea Suspected	Breathing pattern suggests possible apnea event
3	Snoring	Chest vibrations consistent with snoring detected
4	Out of Bed	No person detected on the pad
5	Life Abnormality	Vitals are outside safe range — emergency flag
6	Light Sleep	Person in light sleep phase (rare from 0x0E)

Important: The bridge batches readings and sends one POST request to the backend every 5 seconds (not per frame). The overnight volume is approximately 808 records over ~7 hours for one device.

2. How Data is Stored in the Database

When the backend receives a POST from the bridge, it does two things:

- a) Saves a structured record in the **emfit_logs** collection with a standardized 20-field format (one record per device per 30-second window to avoid duplicates).
- b) Saves the raw payload in the **unknown_route_logs** collection — this is the collection the Dashboard APIs read from.

Fields stored per record

Field	Source	What It Represents
serialnumber	Device	12-digit device serial number (matches device in our DB)
date_occurred	Device timestamp	When the device recorded this reading
in_bed	Derived from status	true if person is on the pad (status 1, 2, 3, 5, or 6)
heart_rate	0x0E frame	Heart rate in beats per minute (30-220 valid range)
respiration_rate	0x0E frame	Breathing rate in breaths per minute
body_movement	0x0E frame	Raw BCG amplitude (0-255) — higher = more vibration
snoring	Derived from status	true when status = 3 (snoring)
restless	Derived from movement	true when body_movement > 100
activity	Same as body_movement	Raw movement value (used for activity chart)
sleep_stage	Derived	'sleeping' if in bed, mapped from status otherwise
glk_status	0x0E frame	Raw status label (in_bed, snoring, out_of_bed, etc.)
glk_sleep_stage	0x4E frame	Sleep stage from device (awake/light/deep/rem/invalid)
battery_level	0x0E frame	Device battery percentage
out_of_bed	Derived from status	true when status = 4
apnea_suspected	Derived from status	true when status = 2

Key point: The dashboard reads from unknown_route_logs. Each record has a 'body' field containing the above data, plus a 'createdAt' timestamp from the server. The dashboard uses the device's own timestamp (date_occurred), NOT the server timestamp, for all time-based calculations — because the server may receive data with a delay.

3. The Sleep Date Window (9 AM to 9 AM)

When a caregiver opens the app and looks at "September 10" sleep data, they expect to see last night's sleep — the night of September 9th into the morning of September 10th.

To make this work, the system uses a **9 AM to 9 AM window**:

Sleep Date "September 10" = all readings from **September 9, 9:00 AM** through **September 10, 8:59 AM** (IST).

This window captures the full night: an early bedtime (say 9 PM on Sep 9) and a late wake-up (say 8 AM on Sep 10) both fall within the same date's window.

The system also fetches 6 extra hours of data on each side of this window (a 'context buffer') so that a sleep session that started just before 9 AM is not accidentally cut off at the boundary.

4. How Readings are Normalized

Before any calculation, the raw database records go through a normalization process that cleans and enriches the data:

Step 1: Filter out non-health records

Device notifications (like 'sleep belt connected') carry no sleep data. These are removed so they don't create phantom data points.

Step 2: Parse each reading into a standard format

Each record becomes a reading with: timestamp, source (glk or sensor), in_bed (true/false), snoring (true/false), restless (true/false), heart_rate, respiration_rate, activity level, and sleep state (sleeping/awake/out_of_bed).

Step 3: Heart rate validation

Heart rate values below 30 or above 220 are discarded as sensor noise. Only readings between 30 and 220 bpm are used for calculations.

Step 4: Snoring cluster smoothing

The GLK device reports snoring status very frequently (~90% of in-bed readings show snoring). A single snoring flag doesn't mean the person was actually snoring — it could be normal breathing vibrations. To filter out false positives, the system uses a **5-minute sliding window**: a reading is marked as 'genuinely snoring' only if at least **3 snoring readings** exist within a 5-minute window around it. Isolated or sporadic flags are ignored.

Step 5: Restless cluster smoothing

Same approach as snoring. The raw 'restless' flag (body_movement > 100) fires on normal breathing micro-movements. A reading is marked 'genuinely restless' only when at least **3 high-movement readings** occur within a **5-minute window**.

Step 6: In-bed debouncing

The sensor can flicker between 'in bed' and 'out of bed' at session edges (e.g., 43 transitions during one night). Brief out-of-bed gaps shorter than 3 minutes between two in-bed readings are filled back in as 'in bed' — they are almost certainly sensor noise, not actual bed exits.

5. Sleep Session Detection

A 'sleep session' represents one continuous period of sleep. The system detects sessions using a state machine with these rules:

Starting a session

A session begins the instant a 'sleeping' state is detected (person lies on the pad and the sensor confirms they are in bed with a sleeping state).

Ending a session

A session ends when the person is confirmed awake or out of bed for **15 continuous minutes**. Brief wake-ups (bathroom visit, getting water) that resolve back into sleep within 15 minutes do NOT split the session — they are absorbed into the session's duration. The session end time is recorded as the moment wakefulness first started, not when the 15-minute timer expires.

Device offline

If the device stops sending data for more than **5 minutes**, any open session is closed at the timestamp of the last known reading. The system does not assume sleep continued through a data gap.

Session merging

If two sessions are less than **60 minutes apart**, they are merged into one. This handles the case where someone wakes briefly (e.g., a 20-minute bathroom trip) and goes back to sleep — the caregiver sees one continuous night, not two fragments.

Which session is displayed

If multiple sessions exist in the same night (e.g., an afternoon nap and the main overnight sleep), the dashboard shows the **most recent (last) session**. This is typically the main overnight sleep.

6. Dashboard Summary API — What Each Number Means

The Summary API powers the main dashboard screen. Here is what each field represents and how it is calculated:

Dashboard Field	How It Is Calculated
is_sleeping	true if the most recent reading in the window shows the person in bed AND in a sleeping state. false if they are out of bed or confirmed awake.
sleep_start	The timestamp when the LAST (most recent) sleep session began. Displayed as a formatted time like '11:30 PM'.
wake_time	The timestamp when the LAST sleep session ended (person woke up). Displayed as a formatted time like '06:15 AM'.
duration_hrs (Sleep Duration)	The duration of the LAST session only, in hours. Calculated as: (wake_time - sleep_start) in hours. Example: 11:30 PM to 6:15 AM = 6.8 hours.
avg_sleep_score_pct (Sleep Score)	Percentage of 30-minute time buckets within the last session where the person was in a 'sleeping' state vs 'awake'. 100% means asleep the entire session. Based on GLK's own sleep classification, not just in-bed presence.
avg_heart_rate	Average heart rate across all 30-minute buckets within the last session. Each bucket averages the HR readings in that 30-min window; the final number is the average of all buckets.
snoring (episodes + total_minutes)	Number of snoring episodes and total snoring duration within the LAST session only. See Section 8 for full snoring calculation logic.
restless (episodes + total_minutes)	Number of restless episodes and total restless duration within the LAST session only. See Section 9 for full restless calculation logic.
activity_intervals	Activity level chart showing movement intensity during the sleep session. Scoped to the last session's time window.
sleep_stage_timeline	Timeline of sleep stages (awake/light/deep/REM/sleep) within the last session. See Section 11.

Key rule: Sleep duration, snoring, restless, heart rate, activity, and sleep stages are ALL scoped to the same last session. This ensures snoring minutes can never exceed sleep duration.

7. Detail API — What Each Number Means

The Detail API powers the expanded sleep detail screen. It includes everything the Summary API has, plus additional breakdowns:

Detail Field	How It Is Calculated
sleep_start / wake_time / duration_hrs	Same as Summary — the LAST session's start, end, and duration.
avg_sleep_score_pct	Same calculation as Summary — percentage of sleeping time within the last session.
avg_sleep_quality	A label derived from the sleep score: 'good' (85%+), 'average' (70-84%), or 'poor' (below 70%).
has_previous_week_data	true if at least one of the previous 7 days had sleep data with a non-zero score. Used by the app to decide whether to show a week comparison.
in_bed (duration_hrs + intervals)	Total time spent in bed (may differ from sleep duration if the person was in bed but awake). Intervals show each continuous in-bed period with start/end times. Brief in-bed gaps under 5 minutes are merged. Clipped to the sleep session boundaries.
heart_rate (avg + 30-min intervals)	Average heart rate and a 30-minute breakdown chart within the last session.
respiration_rate (avg + 30-min intervals)	Average breathing rate and a 30-minute breakdown chart within the last session.
snoring (episodes + total_minutes + episode_details)	Same snoring calculation as Summary, plus a list of each episode with start time, end time, and duration in minutes.
restless (episodes + total_minutes + episode_details)	Same restless calculation as Summary, plus episode details.
activity (intervals)	Activity chart showing movement intensity per interval. Each interval has a 'turns' flag (true if average activity > 5 on a 0-10 scale) and an activity number.
sleep_stage_timeline (intervals)	Timeline of sleep stages within the last session. Each segment has: from time, to time, stage label, and duration in minutes.

8. Snoring Calculation Logic

Snoring calculation involves several steps to produce accurate, realistic numbers:

Step 1: Scope to the last sleep session

Only readings that fall within the last sleep session's start and end time are considered. Earlier sessions, daytime readings, and readings outside the session are excluded. This ensures snoring minutes never exceed sleep duration.

Step 2: Use cluster-smoothed snoring

The system uses the 'snoring_cluster' flag (see Section 4, Step 4) — not the raw snoring flag. A reading counts as snoring only if there were at least 3 snoring readings within a 5-minute window around it.

Step 3: Calculate snoring density

Snoring density = (number of snoring-cluster readings) / (total in-bed readings within the session). This tells us what percentage of in-bed time had snoring. If density is above 70%, it means the sensor is over-reporting snoring (firmware issue), and special capping rules apply.

Step 4: Episode detection

Consecutive snoring readings are grouped into episodes. The rules depend on density:

Rule	Normal Density (under 70%)	High Density (above 70%)
Merge gap	Episodes within 10 minutes of each other are merged into one	Episodes within 1 minute of each other are merged (tighter grouping)
Max episode duration	No limit	Each episode capped at 30 minutes
Max total snoring	No limit	Total snoring capped at 25% of in-bed time
Minimum episode	Must be at least 1 minute	Must be at least 1 minute

Step 5: Total calculation

Total snoring minutes = sum of all episode durations (after merging, capping, and filtering). Each episode has a start time, end time, and duration in minutes.

Why high-density capping exists: The current GLK firmware reports status code 3 ('snoring') as its default in-bed state for ~90% of readings. This is a firmware behavior, not actual snoring. Until GLK updates the firmware, the capping rules prevent the app from showing 'snoring all night' which would be misleading. When the firmware is fixed to report actual snoring (typically 10-30% of the night), the density will be under 70% and the caps will not activate — episodes will flow through normally.

9. Restless / Tossing & Turning Calculation

Restless detection follows the same pattern as snoring:

1. **Scoped to last session** — only readings within the last sleep session.
2. **Cluster-smoothed** — uses 'restless_cluster' flag (3+ high-movement readings in 5 min).
3. **Density-aware capping** — if restless density exceeds 40%:

Rule	Normal (under 40%)	High Density (above 40%)
Merge gap	10 minutes	1 minute
Max episode duration	No limit	20 minutes
Max total restless	No limit	20% of in-bed time
Minimum episode	1 minute	1 minute

Activity chart

The activity chart shows movement intensity during the sleep session on a 0-10 scale. The raw GLK body_movement (0-255) is mapped non-linearly: 0-230 maps to 0-4 (normal baseline, covers ~68% of in-bed readings), and 230-255 maps to 4-10 (actual tossing). A 'turns' flag is set when the average activity in an interval exceeds 5.

10. Heart Rate & Respiration Rate Display

Both vitals are displayed as 30-minute averages within the last sleep session:

1. All valid readings within the session are grouped into 30-minute time buckets.
2. Within each bucket, the average HR and average RR are computed.
3. The overall average is the average of all bucket averages.
4. Only valid readings are used: HR must be between 30-220 bpm; RR must be a positive finite number.

The chart in the app shows one data point per 30-minute bucket, allowing the caregiver to see how heart rate and breathing changed throughout the night.

11. Sleep Stage Timeline

The sleep stage timeline shows how the person's sleep progressed through different stages during the night. It is scoped to the last sleep session.

How stages are determined

The system prefers GLK's own detailed stage classifications (light/deep/rem/awake) when available. If the device only provides binary data (in bed vs. not in bed), the system falls back to a simple 'sleep' vs. 'awake' display.

Timeline segments

Consecutive readings with the same stage are grouped into segments. Each segment has: start time, end time, stage name, and duration in minutes. Adjacent segments of the same stage are merged. Segments shorter than 1 minute are excluded.

Data gaps

If the device stops sending data for more than 5 minutes, the current segment is closed and a new one starts when data resumes. The system does not assume the stage continued during the gap.

Current state: Since the GLK device is not yet producing per-minute light/deep/REM classifications through local TCP (pending GLK team configuration), the timeline currently shows a simplified view. Once the GLK team configures the device, the full stage breakdown will appear automatically — no code changes needed.

12. Vital Alert Notification Logic

The system monitors heart rate and respiration rate in real time and sends push notifications to the caregiver when vitals fall outside safe ranges. Here is the complete logic:

Threshold configuration

Vital	Level	Range	Notification
Heart Rate	Normal	50 – 100 bpm	No alert
Heart Rate	Danger	40 – 49 bpm	"Low Heart Rate" push notification
Heart Rate	Danger	101 – 120 bpm	"Elevated Heart Rate" push notification
Heart Rate	Critical	Below 40 bpm	Critical vital alert (bypasses quiet hours)
Heart Rate	Critical	Above 120 bpm	Critical vital alert (bypasses quiet hours)
Respiration	Normal	12 – 22 brpm	No alert
Respiration	Danger	10 – 11 brpm	"Low Breathing Rate" push notification
Respiration	Danger	23 – 25 brpm	"High Breathing Rate" push notification
Respiration	Critical	Below 10 brpm	Critical vital alert (bypasses quiet hours)

Vital	Level	Range	Notification
Respiration	Critical	Above 25 brpm	Critical vital alert (bypasses quiet hours)

Alert system workflow

Every 60 seconds, the system checks each active GLK device:

- 1. Freshness check** — Is the latest reading less than 10 minutes old? If not, skip (stale data, device may be offline).
- 2. Zero value suppression** — Is HR=0 and RR=0? If yes, skip. Person is off the pad.
- 3. Sensor settling cooldown** — Has the person been on the pad for at least 2 minutes? If they just got on, skip (sensor needs time to calibrate).
- 4. Threshold evaluation** — Check HR and RR against the table above. If both are abnormal, the higher severity wins. HR takes priority in ties.
- 5. Cooldown check** — Has an alert been sent for this device in the last 5 minutes? If yes, skip (prevents alert flooding).
- 6. Direction detection** — Is the value low (below normal minimum) or high (above normal maximum)? This determines the notification type.
- 7. Send notification** — Push notification sent to the caregiver's phone. Critical alerts bypass quiet hours; danger-level alerts respect them.

Medical rationale

Normal resting HR is 60-100 bpm (American Heart Association). During deep sleep, elderly adults may naturally drop to 50-55 bpm — so the lower bound is set at 50 to avoid false alarms while catching clinically significant bradycardia. Normal resting RR is 12-20 brpm, widened to 22 for elderly sleep (REM causes irregular breathing patterns up to 21-22 brpm). The 2-minute settling cooldown prevents false alerts from sensor calibration noise, and the 5-minute alert cooldown prevents alarm fatigue when vitals hover near a threshold.

13. Summary of Key Rules

Rule	Value	Why
Sleep date window	9 AM yesterday to 8:59 AM today	Captures the full overnight sleep in one date's view
Context buffer	6 hours on each side	Prevents sessions near the boundary from being cut off
Awake confirmation	15 minutes	Brief wake-ups don't split the session; sustained waking does
Device offline threshold	5 minutes	If no data for 5 min, session is closed at last known reading
Session merge gap	60 minutes	Sessions less than 60 min apart are merged into one night
Snoring/restless smoothing window	5 minutes, 3+ readings	Filters out isolated sensor noise; keeps genuine episodes
In-bed debounce	3 minutes	Brief out-of-bed flickers under 3 min are treated as sensor noise
HR valid range	30 – 220 bpm	Values outside this are sensor errors, not real heart rates
Snoring scoped to	Last sleep session only	Prevents snoring minutes from exceeding sleep duration
High snoring density cap	25% of in-bed time	Compensates for firmware over-reporting snoring (~90% of readings)
Vital alert check	Every 60 seconds	Near real-time monitoring without overwhelming the server
Vital alert cooldown	5 minutes between alerts	Prevents alarm flooding when vitals hover near a threshold
Sensor settling cooldown	2 minutes after getting on pad	Avoids false alerts from sensor calibration noise

Data flow summary

GLK Sensor (under mattress) → Raspberry Pi Bridge (local WiFi, TCP port 8766) → Backend Server (POST /api/health every 5 seconds) → MongoDB (emfit_logs + unknown_route_logs) → Dashboard Summary API + Detail API (read from unknown_route_logs, normalize, calculate, respond) → Mobile App (displays the numbers)